

DSAA3073: Theories in Data Science

Week 6 Validation Sheet: Optimization & Ensemble Methods: AdaBoost

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Question 1. [5 marks] Which of the following best describes the “weak learning” assumption in boosting?

- a) Each weak learner must achieve at least 90% accuracy on the training set
- b) Each weak learner must perform better than random guessing by some margin $\gamma > 0$
- c) Each weak learner must be a decision tree with depth at most 2
- d) Each weak learner must minimize the exponential loss function

Question 2. [5 marks] In AdaBoost, if a weak hypothesis h_t achieves weighted error $\varepsilon_t = 0.3$ at round t , what is its hypothesis weight α_t ?

- a) $\alpha_t = \ln\left(\frac{7}{3}\right) \approx 0.847$
- b) $\alpha_t = \left(\frac{1}{2}\right) \ln\left(\frac{7}{3}\right) \approx 0.424$
- c) $\alpha_t = \ln\left(\frac{3}{7}\right) \approx -0.847$
- d) $\alpha_t = \left(\frac{1}{2}\right) \ln\left(\frac{3}{7}\right) \approx -0.424$

Question 3. [5 marks] How does the weight $D_{t+1}(i)$ of a correctly classified example compare to its weight $D_t(i)$ in AdaBoost?

- a) It increases by a factor of e^{α_t}
- b) It decreases by a factor of e^{α_t}
- c) It remains unchanged
- d) It is set to zero

Question 4. [5 marks] If all weak learners have edge at least γ , which bound correctly describes AdaBoost's training error after T rounds?

- a) Training error $\leq e^{-2\gamma^2 T}$
- b) Training error $\leq e^{-\gamma T}$
- c) Training error $\leq (1 - 4\gamma^2)^T$
- d) Training error $\leq T \cdot e^{-\gamma^2}$

Question 5. [5 marks] Which statement about the exponential loss used in AdaBoost is FALSE?

- a) It upper bounds the 0-1 loss
- b) It is convex and differentiable
- c) It equally penalizes all misclassifications regardless of confidence
- d) It can lead to overfitting on noisy data

Question 6. [5 marks] What is the role of the normalization factor Z_t in AdaBoost?

- a) It ensures all example weights sum to 1 after each round
- b) It controls the learning rate of the algorithm
- c) It measures the quality of the weak learner h_t
- d) It determines when to stop boosting

Question 7. [5 marks] Compared to bagging, which characteristic is most distinctive to boosting algorithms like AdaBoost?

- a) Using multiple weak learners
- b) Training learners on weighted or resampled data
- c) Sequential training where each learner focuses on previous errors
- d) Aggregating predictions through voting

Question 8. [5 marks] For a weak learner with weighted error $\varepsilon_t = 0.4$, what happens to its hypothesis weight α_t ?

- a) $\alpha_t > 1$
- b) $0 < \alpha_t < 1$
- c) $\alpha_t = 0$
- d) $\alpha_t < 0$

Part B: Mixed Format Questions (90 points)

Question 9. [15 marks] Consider the normalization factor in AdaBoost: $Z_t = \sum_{i=1}^n D_t(i) \cdot e^{-\alpha_t y_i h_t(x_i)}$.

(a) [8 marks] Express Z_t in terms of the weighted error ε_t and the hypothesis weight α_t . Show your derivation by separating the sum over correctly and incorrectly classified examples.

(b) [7 marks] Show that $Z_t = 2\sqrt{\varepsilon_t(1 - \varepsilon_t)}$ when $\alpha_t = \frac{1}{2} \ln \frac{1 - \varepsilon_t}{\varepsilon_t}$.

Question 10. [13 marks] Consider a binary classification problem with 5 training examples. The table shows the true labels and the predictions of three weak learners:

Example	y_i	$h_1(x_i)$	$h_2(x_i)$	$h_3(x_i)$
1	+1	+1	+1	-1
2	+1	-1	+1	+1
3	-1	-1	+1	-1
4	-1	-1	-1	-1
5	+1	+1	-1	+1

- (a) [5 marks] Compute the hypothesis weight α_1 for h_1 assuming uniform initial weights $D_1(i) = \frac{1}{5}$.
- (b) [8 marks] Compute the updated weights $D_2(i)$ after incorporating h_1 . Verify they sum to 1.

Question 11. [15 marks] Prove that the training error of the final AdaBoost hypothesis $H(x) = \text{sign}\left(\sum_{t=1}^T \alpha_t h_t(x)\right)$ satisfies:

$$\hat{R}(H) \leq \prod_{t=1}^T Z_t \quad (9)$$

where Z_t is the normalization factor at round t .

Question 12. [12 marks] Suppose all weak learners satisfy $\varepsilon_t \leq \frac{1}{2} - \gamma$ for some fixed $\gamma > 0$.

(a) [6 marks] Show that $Z_t \leq \sqrt{1 - 4\gamma^2}$ for all t .

(b) [6 marks] Use this to prove that after T rounds, the training error satisfies:

$$\hat{R}(H) \leq e^{-2\gamma^2 T} \quad (18)$$

Question 13. [10 marks] AdaBoost can be viewed as performing coordinate descent on the exponential loss.

(a) [5 marks] Write the exponential loss over the training set as a function of the ensemble prediction

$$f(x) = \sum_{t=1}^T \alpha_t h_t(x).$$

(b) [5 marks] Explain how the greedy selection of h_{t+1} and α_{t+1} at each round corresponds to a coordinate descent step. What is the “coordinate” being optimized?

Question 14. [15 marks] Compare AdaBoost with bagging in terms of:

- (a) [5 marks] How training data is presented to each weak learner
- (b) [5 marks] Variance reduction vs. bias reduction
- (c) [5 marks] Sensitivity to noisy labels and outliers

